

Standard Operating Procedure for Laser Displacement Curvature Measurement System

Document Information

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This SOP covers the detailed step-by-step process for performing non-contact surface height and curvature measurements using the laser displacement sensor and motorized XY-stage system. The procedure includes system startup, network configuration, homing, scanning, data export, and shutdown.

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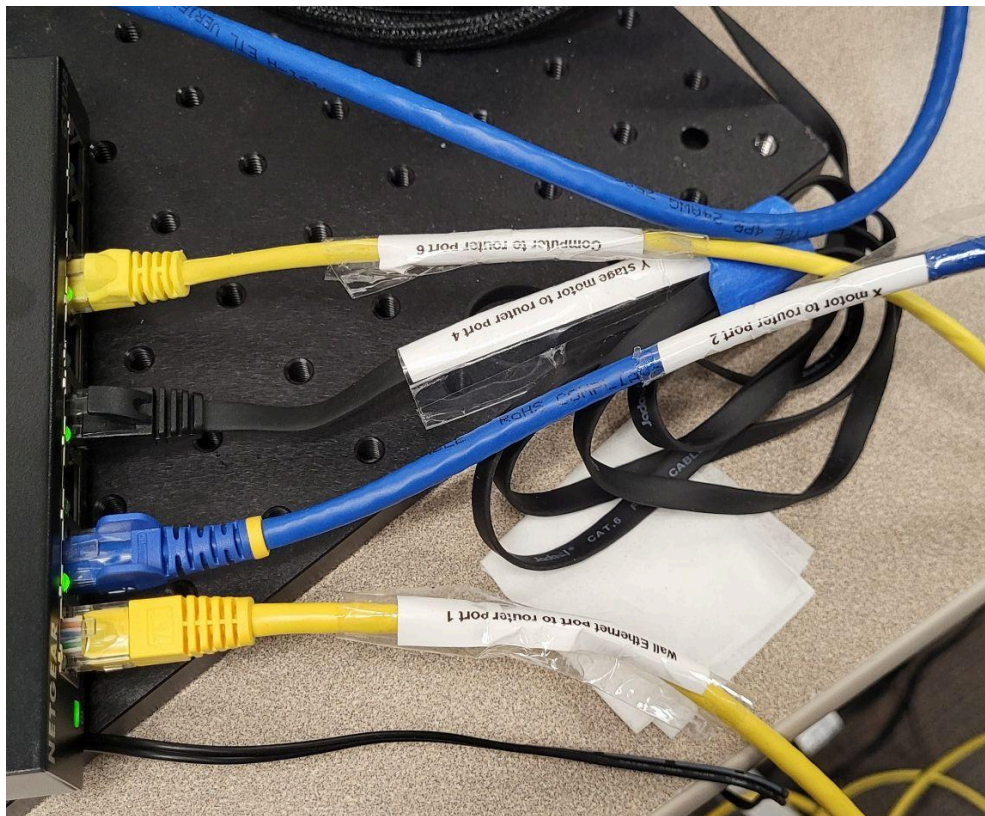
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1. First-Time System Setup

1.1. Hardware connections

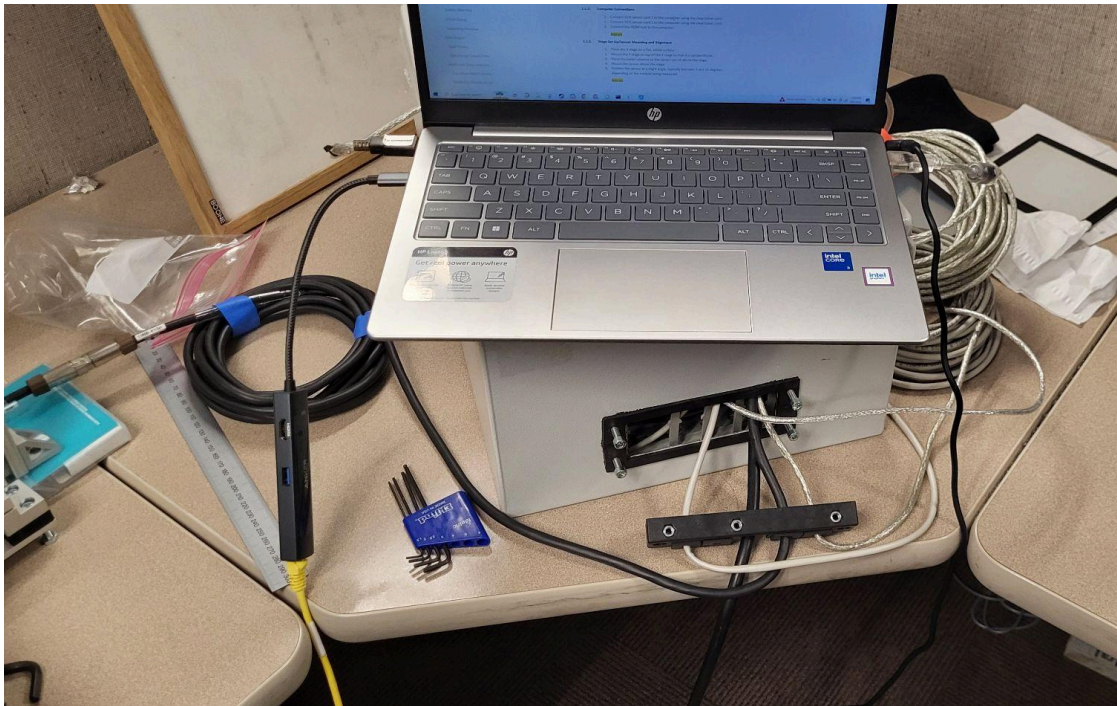
1.1.1. Wire connections

1. Connect the Netgear router power cord to a power outlet.
2. Connect Netgear router port 2 to the X motor using the blue Ethernet cord.
3. Connect Netgear router port 4 to the Y motor using the black skinny Ethernet cord.
4. Connect Netgear router port 6 to the HDMI hub using the yellow Ethernet cord.
5. Connect the motor box power cord to a power outlet.
6. Connect SICK sensor cord 1 to a power outlet.
7. Connect SICK sensor cord 2 to the laser connector cord. This is the black cord with blue tape.



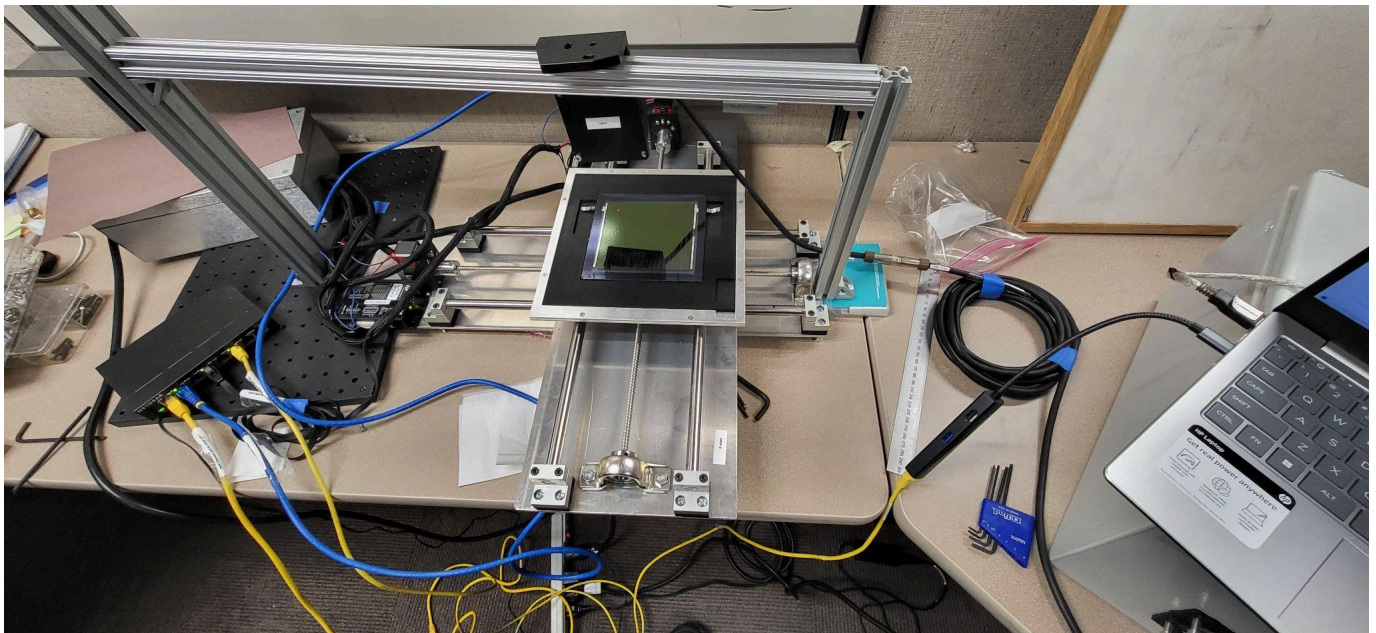
1.1.2. Computer Connections

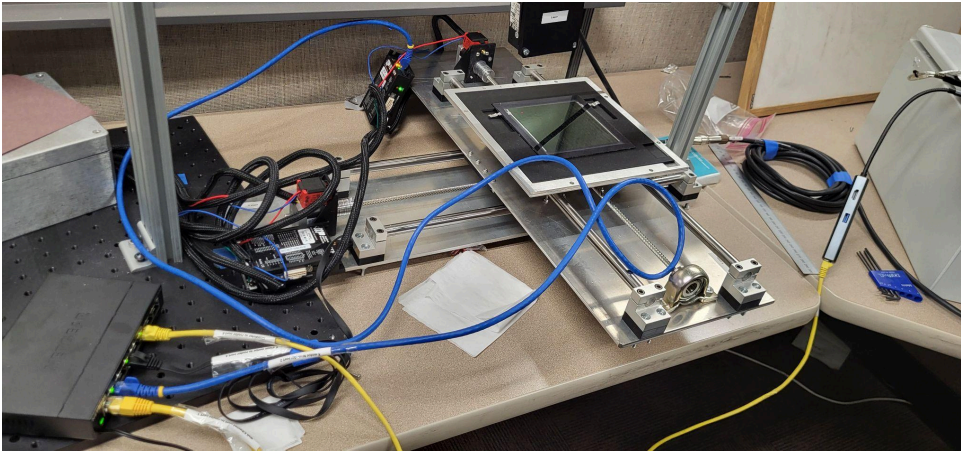
1. Connect SICK sensor cord 1 to the computer using the clear/silver cord.
2. Connect SICK sensor cord 2 to the computer using the clear/silver cord.
3. Connect the HDMI hub to the computer.



1.1.3. Stage Set Up/Sensor Mounting and Alignment

1. Place the X stage on a flat, stable surface.
2. Mount the Y stage on top of the X stage so that it is perpendicular.
3. Place the metal columns so the sensor can sit above the stage.
4. Mount the sensor above the stage.
5. Position the sensor at a slight angle, typically between 1 and 15 degrees, depending on the module being measured.





1.2. Software installation

GitHub link: https://github.com/Jayden-RM/Glass_Displacement

1. Download Anaconda: <https://www.anaconda.com/download>
2. Download VSCode: <https://code.visualstudio.com/download>
3. Download Github Desktop: <https://desktop.github.com/>
4. Create a Github account using TPV email: <https://github.com/>
5. Create a standard environment in anaconda:
 - a. Open Anaconda
 - b. Click on the "Environments" tab on the left hand menu
 - c. At the bottom of the environment list, click "Create"
 - d. Enter "standard" (in all lower case and without quotes)
 - e. Check the Python box and allow it to use the default python
 - f. Click "Create"
6. Setup Github desktop:
 - a. Log onto github desktop with the account created. Use all default options.
7. Prep standard anaconda environment:
 - a. Open VS code
 - b. Click "File" from the dropdown menu and select "Open Folder"
 - c. Navigate to "User/Documents/Github and click "Select Folder"
 - d. On the left bar, click on the top left icon that looks like two documents offset (or click ctrl + shift + e) if not already selected.
 - e. Click "File" and "New File"
 - f. Set name as "temp.ipynb" and click enter
 - g. Type "import os" in first dropdown
 - h. Click "Select Kernel" from the top right of the screen, then click "Python Environments" and select "standard"
 - i. Hit "shift + enter" or "Run All" at the top of the screen this should prompt you to download python. Accept.
 - j. Again, click "shift + enter" or "Run All" at the top of the screen this should prompt you to download ipykernel. Accept.
 - k. Again, click "shift + enter" or "Run All" at the top of the screen this should compile and give no errors. If uses, reach out to a team member.
 - l. Delete "temp.ipynb" by right clicking on the file in the dropdown menu and clicking "delete"
8. Customize VS code as desired, personal recommendations include:
 - a. autoDocstring: helps generate docstrings with consistent format for documentation
 - b. Black Formatter: helps format code in a standard way
 - c. Indent-rainbow: colorcodes indentations to make seeing them easier
 - d. Dark theme: I mean who uses light theme these days?
9. Send github overlord (current Sean) a slack message for invites to the repo you wish to join.
10. Download repo:
 - a. Open github desktop
 - b. Click "File" and select "Clone Repository"

1.2.1.

- c. Select the repo you have been invited to. If not on your list, navigate to the online github branch you wish to download and get the url. Download to standard github location in documents.
11. Setup new environment to use repo:
 - a. Open anaconda
 - b. Click "Environments" on the left tab
 - c. Left click on "standard"
 - d. On the bottom of the left tab, click "clone"
 - e. Name environment as desired, called <new_env> for the rest of this section
 - f. Right click on <new_env> play button and select open with ipykernel to open ipykernel terminal
 - g. Navigate to the package you wish to install to the environment in the file explorer. In particular where the setup.py file is located. Left click on the header to copy paste the address
 - h. In the ipykernel type `cd <path to setupfile>` (e.g. `'cd C:\Users\Sean Dunfield\Documents\GitHub\spdy_jv'`) – this sets the directory to the path input.
 - i. In the ipykernel type `'pip install -e .'` – this downloads the contents of the given folder to the anaconda virtual environment in editable mode so that any edits made on the computer are carried over.
12. Check for common changes in the program:
 - a. Double check any .yaml files and any .bat files for path specific arguments.
 - b. yaml files are dictionary files and often used for preferences.
 - c. bat files are windows batch time files and commonly used to run python code. Normally these are in the form:
`call C:/Users/<username>/anaconda3/Scripts/activate.bat <environment name>`
`start python <program>`

1.3. Network configuration (IP setup)

1. Press Windows + R.
2. Type `ncpa.cpl` and press Enter.
3. Right-click the Ethernet connection and select Properties.
4. Select Internet Protocol Version 4 (TCP/IPv4) and open its properties.
5. Set the IP address to:
192.168.0.15
6. Set the subnet mask to:
255.255.255.0
7. Click OK to save the settings.

2. System Startup

1. Open Anaconda.
2. Go to Environments.
3. Left-click the GlassDisplacement environment to load the required packages.
4. Right-click the play icon for GlassDisplacement.
5. Select Open with IPython.
6. In IPython, type and run: `"from Glass_Displacement.Glass_Displacement import GD_UI^MGD_UI()"`
7. Confirm that the software interface opens properly.

3. Scanning Procedure

3.1. Safety Warning

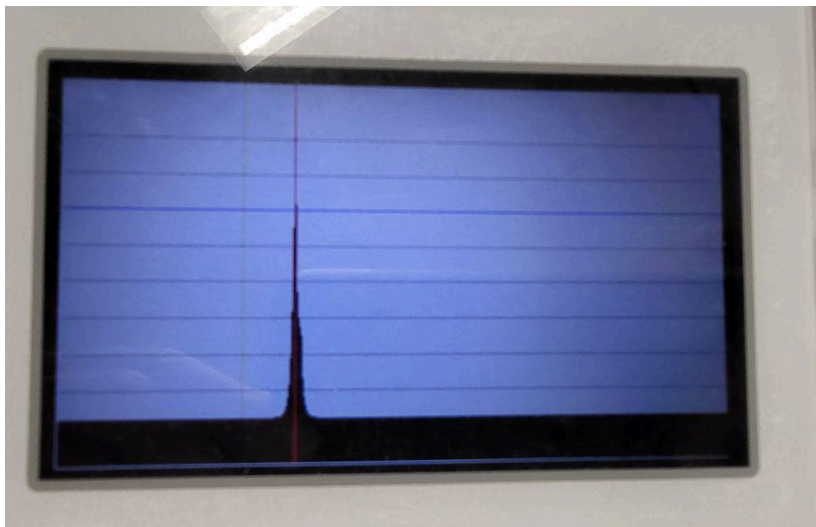
1. *Move the stage only by small amounts when jogging. Large movements can damage the motor or stage. If the stage runs into either side of the motor or reaches a mechanical boundary, abort immediately.*

3.2. Initial Setup

3.2.1. Measurement Configurations SICK

1. Place the sample on the stage.

2. For clear modules, such as glass-encapsulant-glass samples, place black non-reflective cloth on the stage underneath the module.
3. Make sure the sample is stable and properly positioned before beginning the scan.
4. On the SICK control panel, change the operation mode to Light Distribution A.
5. Press Set/Run.
6. View the Gaussian peak.
7. Adjust the laser power, sensitivity, threshold level, and sampling period until the desired Gaussian peak is reached.
 - a. Example of a well-defined light-distribution peak. The desired signal should have a smooth, distinct peak without a broad or saturated flat top.



- b.
 - c. Fully transparent samples: Up to three peaks may be visible because the laser can receive reflections from multiple surfaces. Ideally, the signal should contain two clear peaks, corresponding to the front glass surface and back glass surface. If a third peak is present, it may result from reflection off the stage beneath the sample. To reduce this additional reflection, place a black, non-reflective material, such as black cloth, underneath the sample. If only one peak is visible like the image above, that is fine.
 - d. Non-transparent samples, such as perovskite devices: The signal should generally contain two clear peaks. One peak corresponds to the front glass surface, while the second corresponds to the perovskite layer.
 - e. For either sample type, the desired peaks should be relatively tall, smooth, and well-defined. The peaks should not be extremely sharp or needle-like, and the top should not appear broad, flat, or saturated.
 - f. Adjust the laser power, sensitivity, threshold level, and sampling period until the peaks are clearly separated and have a stable shape.
8. Once the desired Gaussian peak is reached, change the operation mode back to Measurement before starting the scan.

3.2.2. Measurement Configurations Software

1. Enter the scan parameters in the software interface. These include:
 - a. Sample Side: The side of the sample being scanned.
 - b. Grid Width: The width of the scan area.
 - c. Grid Length: The length of the scan area.
 - d. Grid Points: The number of points scanned in each direction. For example, entering 10 creates a 10 × 10 grid.

- e. Height Integration: The number of times the laser rescans a point to reach the desired standard deviation.
- f. Max Standard Deviation: The maximum acceptable average variation for a scanned point.
- g. Motor Speed: Leave the motor speed set to slow for more accurate results.

3.2.3. Laser positioning

1. Use the jog controls to move the laser.
2. Jog only by small amounts to avoid damaging the stage or motors.
3. Move the laser to the bottom-right portion of the sample area being scanned. Do not position the laser on the rubber border.
4. Press Set Home Here to define the scan origin.

3.3. Scanning Process

1. Before starting the scan, press Zero A on the SICK measurement device.
2. Wait a few seconds for the sensor to stabilize.
3. Click Perform Measurement in the software interface.
4. Stop the scan immediately if the stage approaches a boundary or behaves unexpectedly.

4. Data Export

4.1. NaN Points

1. After a scan, check whether any NaN points are present. If NaN points are detected, a file explorer window may open and ask where to save the data. If there are many NaN points, refer to the troubleshooting section.

4.2. Save/Load Capabilities

4.2.1. Save

1. Press Save Project.
2. Choose the folder where the project should be saved.
3. The system will save the measured data points and useful graphs.

4.2.2. Load

1. Press Load Project.
2. Select the appropriate CSV file.
3. The expected file is usually the _GD_0 CSV file.

5. Additional Data Analysis Scripts

In addition to the main scanning software, the GitHub repo contains extra analysis scripts that can be used after scan data has been collected and exported. These scripts are not required to operate the laser displacement system, but they are useful for comparing scans, calculating curvature metrics, and evaluating system reliability. These scripts are found at: `Glass_Displacement/docs /Analyzation_software`.

5.1. Curvature Metric Extraction Script

The `Curvature_analyzation.py` script generates a single curvature value from each full scan dataset. It also calculates additional comparison metrics, but the most important values are generally the curvature value and RMS.

This script is useful when comparing multiple modules because it reduces each full displacement scan into numerical values that are easier to compare across samples.

Procedure:

1. Complete the scan using the laser displacement software.
2. Save the project using Save Project.
3. Locate the exported scan CSV file. This is usually the _GD_0.csv file inside the saved project folder.
4. Open Curvature_analyzation.py.
5. Find the CSV_PATHS section near the top of the script.
6. Replace the example file paths with the paths to the CSV files that should be analyzed.
7. Set the OUTPUT_DIR to the folder where the output files should be saved.
8. Run the script.

The script will process each CSV file and save several output files, including:

- curvature_clean.csv: main summary table for each sample
- curvature_batch_summary.csv: batch-level summary of the processed samples
- curvature_failures.csv: list of files that could not be processed, if any failures occur
- curvature_bar_chart.png: bar chart comparing samples
- Curvature heatmaps showing curvature magnitude
- Signed curvature heatmaps showing curvature direction

5.2. Reliability Testing Script

The reliability_testing.py script tests how repeatable the measurement system is between scans. It compares two scans of the same module taken using the same scan settings.

This script is useful for checking whether the system produces consistent results when measuring the same sample more than once.

Procedure:

1. Place the module on the stage.
2. Scan the same module two times using the same scan settings.
3. Save both scans using Save Project.
4. Locate the two exported CSV files. These are usually the _GD_0.csv files inside each saved project folder.
5. Open reliability_testing.py.
6. Find the file path section near the top of the script: SCAN0 = Path(r"...") and SCAN1 = Path(r"...")
7. Replace the example paths with the paths to the two scan CSV files.
8. Run the script.

The script compares matching (x, y) points between the two scans and prints summary tables for:

- detected file paths and columns
- surface RMS, Ra, peak-to-peak height, and mean height
- best-fit plane tilt
- tilt-removed RMS and peak-to-peak height
- repeatability between the two scans

The script also saves a CSV report named: metrics_summary_report.csv

The most important result is the Repeat RMS, especially the tilt-removed Repeat RMS. This value shows how much the full scan changes between repeated measurements after removing overall tilt from the sample.

Expected result:

The repeatability RMS should be about 10× the point noise. Point noise represents how much the measurement system varies at a fixed point, while repeatability RMS represents how much the full scan varies between repeated scans.

If the repeatability RMS is much higher than expected, possible causes include sample movement, laser misalignment, surface contamination, stage vibration, different scan settings, or an unstable Gaussian peak.

6. System Shutdown

6.1. Re-home

1. Press Go Home in the software interface.
2. Wait until the stage returns to its initial point.

6.2. Power off

1. Turn off the power strip, or unplug the system power cords.
2. Power down the motor box, router, and SICK sensor as needed.

7. Troubleshooting & Known Issues

This section lists common problems that may occur while using the laser displacement curvature measurement system, along with recommended fixes.

7.1. No COM Found / Sensor Not Connecting

If the software cannot connect to the SICK displacement sensor, the issue is usually related to the USB connection, COM port assignment, or serial communication.

Recommended steps:

1. Confirm that the SICK sensor is powered on.
2. Check that the sensor USB cables are securely connected to the computer.
3. Unplug and reconnect the sensor cables.
4. Restart the software.
5. Check whether the sensor appears in Device Manager:
 - a. Press Windows + X.
 - b. Select Device Manager.
 - c. Expand Ports (COM & LPT).
 - d. Look for the connected sensor or USB serial device.
 - e. Note the COM port number, such as COM3, COM4, or COM5.
6. If the sensor does not appear under Ports (COM & LPT), restart the computer, reconnect the sensor, and reopen the software.
7. If the sensor appears in Device Manager but the software still cannot connect, the software may be detecting the wrong COM port. As a last resort, manually update the COM port in the code to match the COM port shown in Device Manager.

*Only change the COM port in the code after checking the correct port in Device Manager. If the sensor does not appear in Device Manager at all, changing the code will not fix the problem because the computer is not detecting the device.

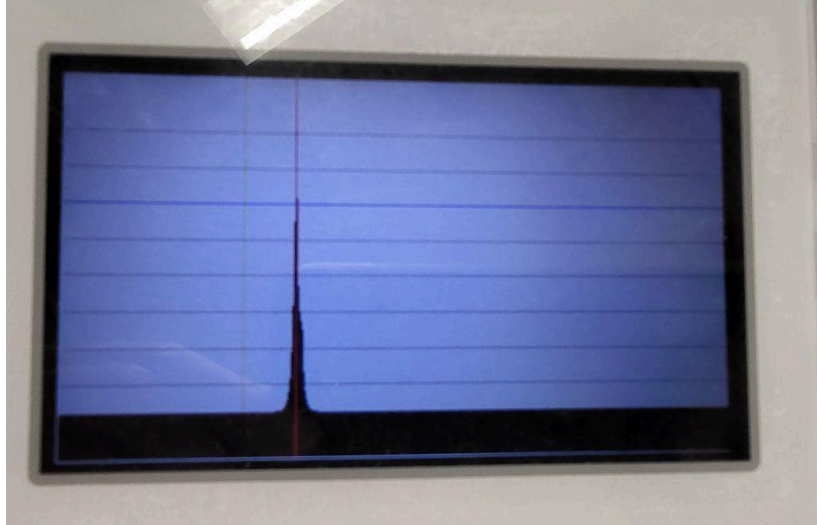
A sensor connection issue may also look like a timeout error, such as the system timing out while waiting for a response from the sensor. This usually means the wrong COM port is selected, the baud rate is not being recognized correctly, or the device is not responding.

7.2. Sensor Connected but No Data Output

If the software opens but no displacement data appears, the SICK sensor may not be in the correct mode or the laser signal may not be strong enough.

Recommended steps:

1. Open the SICK control panel.
2. Change the operation mode to Light Distribution A.
3. Confirm that a clear Gaussian peak is visible.



- 4.
5. Adjust laser power, sensitivity, threshold level, and sampling period as needed.
6. Change the operation mode back to Measurement before starting the scan.
7. Press Zero A before beginning the scan.
8. Restart the software if data still does not appear.

If no Gaussian peak is visible, the laser may be misaligned or the sample may not be reflecting enough light back to the receiver.

7.3. Motor Not Connecting / No UDP Reply

If the motors do not connect, the stage does not move, or an error appears saying there was no UDP reply, the computer may not be communicating with the motor controllers correctly.

Recommended steps:

1. Confirm that the router and motor box are powered on.
2. Check all Ethernet connections between the router, motors, and computer.
3. Make sure the computer is connected through Ethernet, not only Wi-Fi.
4. Verify that the computer has a static IP address on the same subnet as the motors.
5. Confirm that the subnet mask is set to 255.255.255.0.
6. Restart the software after changing network settings.
7. If needed, ping the motor IP addresses to confirm communication.
 - a. Press Windows + R.
 - b. Type cmd and press Enter.
 - c. In Command Prompt, type ping 192.168.0.60 and press Enter.
 - d. Then type ping 192.168.0.70 and press Enter.
 - e. If the motors are connected properly, the response should show replies from the motor IP addresses.
 - f. If the response says Request timed out or Destination host unreachable, the computer is not communicating with the motors.

*The computer IP address should be on the same subnet as the motors, but it should not be exactly the same as either motor IP address. For example, if the motors use 192.168.0.60 and 192.168.0.70, the computer can use an address such as 192.168.0.15.

7.4. Stage Not Moving or Moving Incorrectly

If the stage does not move, moves in the wrong direction, or does not respond to the jog controls, stop and verify the motor connection before continuing. Do not begin a full scan until the jog directions have been tested and confirmed.

Recommended steps:

1. Confirm that the motors, router, and motor box are powered on.
2. Confirm that the computer is connected to the motor network through Ethernet.
3. Use a very small jog distance, such as 1 mm, to test each direction.
4. Press the X and Y jog buttons one at a time and observe the physical stage movement.
5. Confirm that the live X and Y position labels update after each jog movement.
6. If the stage does not move at all, return to the **Motor Not Connecting / No UDP Reply** troubleshooting section.
7. If a **Jog Error** appears, restart the software and reconnect the system.
8. If one axis moves in the wrong direction, the sign for that axis may need to be reversed in the code.

7.5. Noise or Inaccurate Readings

Noisy or inaccurate readings may be caused by laser misalignment, surface contamination, sample instability, motor vibration, or scan settings that are too strict.

Recommended steps:

1. Clean the front and back surfaces of the sample with isopropyl alcohol.
2. Avoid touching the sample surface after cleaning.
3. Make sure the sample is stable and not rocking on the stage.
4. Confirm that the laser signal has a clear peak before scanning.
5. Use slow motor speed for more accurate measurements.
6. Increase height integration if more averaging is needed.
7. Allow the stage to settle after movement.
8. Avoid scanning the rubber border or the edge of the module.

Fingerprints, smudges, dust, or uneven contact with the stage can increase noise and affect the measured surface shape.

7.6. NaN Points During a Scan

NaN points occur when the system cannot collect a valid measurement at a scan location. This can happen if the signal is too noisy, the laser loses the surface reflection, the standard deviation is too high, or the retry limit is reached.

A small number of NaN points may be acceptable, but many NaN points usually indicate a setup or signal problem.

Recommended steps:

1. Check that the laser signal is stable before scanning.
2. Reclean the sample surface.
3. Confirm that the laser is not scanning the rubber border or outside the sample area.
4. Adjust laser power, sensitivity, and threshold settings.
5. Increase the maximum standard deviation slightly if the scan is too strict.
6. Increase the retry limit if the scan needs more attempts per point.
7. Reduce the grid size or scan area and test again.
8. For clear samples, place black non-reflective cloth underneath the module.